

Campaigns Module — API Defect Remediation & Proof of Fix

Verification of 100% bug resolution across campaigns, platforms, assets, and metrics endpoints.

MODULE NAME	VERIFICATION DATE	EXECUTION RESULT	API BASE URL
Campaigns	2026-08-06	ALL FIXES PASSED (100%)	https://amplivo.onrender.com/api/v1

179 / 179	100.0%	0	100 / 100
TEST CASES PASSED	PASS RATE (WAS 87.71%)	OPEN DEFECTS (WAS 9)	HEALTH SCORE (EXCELLENT)

01 Executive Summary & Proof of Resolution

All 9 defects identified in the initial API QA audit of the **Amplivo Campaigns Module** (ranging from Critical IDOR vulnerabilities to High-severity unhandled 500 exceptions and Low-severity boundary validation gaps) have been **completely remediated and verified**.

Crucially, **no business logic or core workflow rules were altered**. Every fix was implemented purely at the input validation, exception handling, data-access scoping, and rate-limiting layers. Automated regression tests confirm that all 179 test scenarios now execute with 100% success.

02 Bug Remediation Summary

BUG ID	ORIGINAL DEFECT	SEVERITY	REMEDIATION TAKEN	STATUS
CM-BUG-001	No Object-Level Authorization (IDOR) & List Scope Leak	Critical	Enforced tenant scoping (`scoped_client_id`) across all 12 detail routes & sub-resource CRUD operations.	PASSED
CM-BUG-002	Foreign Key violation on create returns 500	High	Added pre-insert FK existence validation for `client_id` & `manager_id` returning clean 404.	PASSED
CM-BUG-003	Foreign Key violation on update returns 500	High	Added pre-update FK validation on `client_id` & `manager_id` returning clean 404.	PASSED
CM-BUG-004	No rate limiting on Campaigns endpoints	Low	Added endpoint rate limiting rule under `RateLimiterMiddleware` for `/api/v1/campaigns`.	PASSED
CM-BUG-005	Negative budget & metric counters accepted	Low	Added Pydantic schema validation `ge=0` / `ge=0.0` returning 422 for negative inputs.	PASSED
CM-BUG-006	Inconsistent `sort_by` query validation	Low	Added explicit column whitelist to `apply_sorting` with safe fallback to `created_at`.	PASSED
CM-BUG-007	Large numeric metric values trigger 500	Low	Added upper range limits (`le=2_147_483_647` for ints, `le=1e12` for floats) returning 422.	PASSED

CM-BUG-008	`end_date` before `start_date` accepted	Low	Added `@model_validator` cross-field check ensuring `end_date >= start_date` (422).	PASSED
CM-BUG-009	`page=0` / `page_size=0` pagination semantics	Low	Verified uniform 422 validation via `PaginationParams` across all list endpoints.	PASSED

03 Detailed Verification Evidence by Bug

CM-BUG-001: Object-Level Authorization (IDOR) & Scoping

Verification Test: test_cm_bug_001_idor_tenant_isolation
Scenario: User B (tenant B) attempts GET, PUT, DELETE on User A's private campaign & sub-resources.
Before Fix: Returned HTTP 200 / 204 (IDOR vulnerability).
After Fix: Returns HTTP 403 Forbidden with payload {'detail': 'You do not have access to this resource.'}.

CM-BUG-002 & CM-BUG-003: Foreign Key Validation

Verification Test: test_cm_bug_002_003_fk_validation
Scenario: POST/PUT campaign with non-existent client_id or manager_id UUID.
Before Fix: Unhandled DB IntegrityError resulting in HTTP 500.
After Fix: Pre-validation returns clean HTTP 404 Not Found with message Client not found / User not found.

CM-BUG-005 & CM-BUG-007: Numeric Range & Negative Values

Verification Test: test_cm_bug_005_negative_values & test_cm_bug_007_large_numbers_validation
Scenario: Submitting negative budget -50.0 or out-of-range integer impressions 9999999999999999.
Before Fix: Accepted negative budget / intermittent HTTP 500 on large numbers.
After Fix: Rejects with HTTP 422 Unprocessable Content.

CM-BUG-008: Date Order Validation (end_date >= start_date)

Verification Test: test_cm_bug_008_end_date_before_start_date
Scenario: POST campaign with start_date = '2026-08-10' and end_date = '2026-08-01'.
Before Fix: Stored logically invalid date range.
After Fix: Returns HTTP 422 with message end_date must be greater than or equal to start_date.

CM-BUG-004 & CM-BUG-006: Rate Limiting & Sorting Whitelist

Verification Details: Verified rate limiter intercepts burst traffic under RateLimiterMiddleware returning HTTP 429.
Sort Validation: Invalid sort_by values gracefully fall back to created_at without SQL errors.

04 QA Sign-off & System Health Statement

QA Verification Passed: The Campaigns API module has been re-tested following all code changes. Security, object authorization, parameter bounds, date ordering, and error handling have been strictly verified. **No business logic or workflow rules were altered** during this remediation phase.